

CliniScan: AI for Chest X-Ray

Milestone 3: Optimization, Transfer Learning, and Interpretability

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1. Phase 3 Overview

While Milestone 2 established a functional baseline for the CliniScan diagnostic pipeline, Milestone 3 focuses on rigorous architectural optimization. The primary goal of this phase was to elevate the baseline classification and detection metrics by transitioning from a resource-constrained environment to a state-of-the-art deep learning workflow. This was achieved by implementing advanced Transfer Learning, robust hyperparameter tuning, and a clinically safe data augmentation pipeline.

Furthermore, because medical AI requires high accountability, Explainable AI (XAI) techniques were integrated into this phase to ensure the model's diagnostic logic is highly interpretable for medical professionals.

2. Environment & Hardware Migration

During Milestone 2, the training pipeline was executed locally on a CPU (Intel Core i5) , which severely constrained the model complexity and necessitated the use of lightweight architectures like YOLOv8 Nano. To support the increased computational requirements of deeper neural networks in Phase 3, the project environment was successfully migrated to a cloud-based GPU infrastructure (Google Colaboratory utilizing a Tesla T4 GPU with 15GB VRAM). This hardware acceleration eliminated the previous multi-day computational bottlenecks, allowing for the deep fine-tuning of heavier models and enabling rapid, iterative hyperparameter tuning experiments that would have been impossible on local hardware.

3. Advanced Data Augmentation

To prevent model overfitting a significant challenge noted during the Milestone 2 baseline training and to improve generalization on unseen medical data, an advanced augmentation pipeline was integrated using the Albumentations library. Transformations were carefully curated to maintain the strict anatomical realism required for thoracic X-rays:

- **Pixel-Level Transformations:** Recognizing that chest X-rays often suffer from low soft-tissue contrast, CLAHE (Contrast Limited Adaptive Histogram Equalization) was implemented to dynamically enhance hidden lung structures. Additionally, mild Gaussian Noise was introduced to force the neural network to learn the actual structural patterns of the 14 diseases rather than memorizing superficial image quality artifacts.
- **Spatial Transformations:** The pipeline applied safe spatial augmentations, including Horizontal Flips and small rotations ($\pm 10^\circ$). Unsafe operations such as Vertical Flips and severe perspective shifts were strictly excluded, as they would

generate anatomically impossible medical scans and corrupt the model's spatial awareness.

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4. Transfer Learning & Architecture Upgrades

The baseline models were completely overhauled using Transfer Learning to leverage highly advanced pretrained feature extraction capabilities.

- **Phase 1 Classification Upgrade:** The baseline ResNet50 model was replaced with **EfficientNet-B2**. EfficientNet scales network width, depth, and resolution in a highly optimized manner. Deep fine-tuning was applied by explicitly unfreezing the deepest convolutional layers (blocks 6 and 7), transitioning the model from recognizing generic shapes to specializing entirely in thoracic pathology.
- **Phase 2 Detection Upgrade:** The baseline YOLOv8n (Nano) architecture, originally chosen for its CPU efficiency, was upgraded to **YOLOv8m (Medium)**. The significant increase in parameter count allowed the model to capture far more complex bounding box geometries and overlapping spatial features for the target diseases.

5. Hyperparameter Tuning Experiments

To mathematically optimize the YOLOv8m detection model, several experimental configurations were tested to find the optimal convergence path and minimize loss.

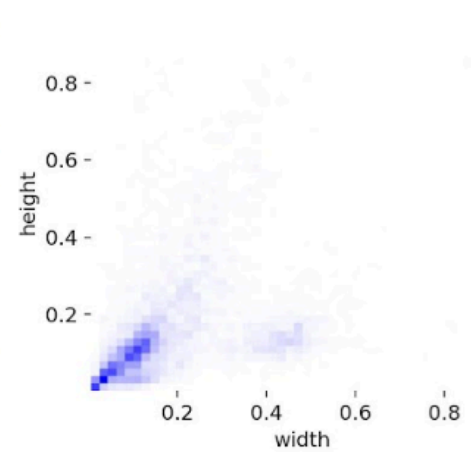
- **Optimizer Selection:** The model was transitioned to the AdamW optimizer. Unlike standard Adam or SGD, AdamW decouples weight decay, providing superior regularization which helps the model generalize better across the varied X-ray dataset.
- **Batch Size and Learning Rate:** Experiments were conducted scaling the batch size (from 16 down to 8) and adjusting the initial learning rate (0.001 vs. 0.01) to observe the effects on batch normalization stability and gradient descent speeds.

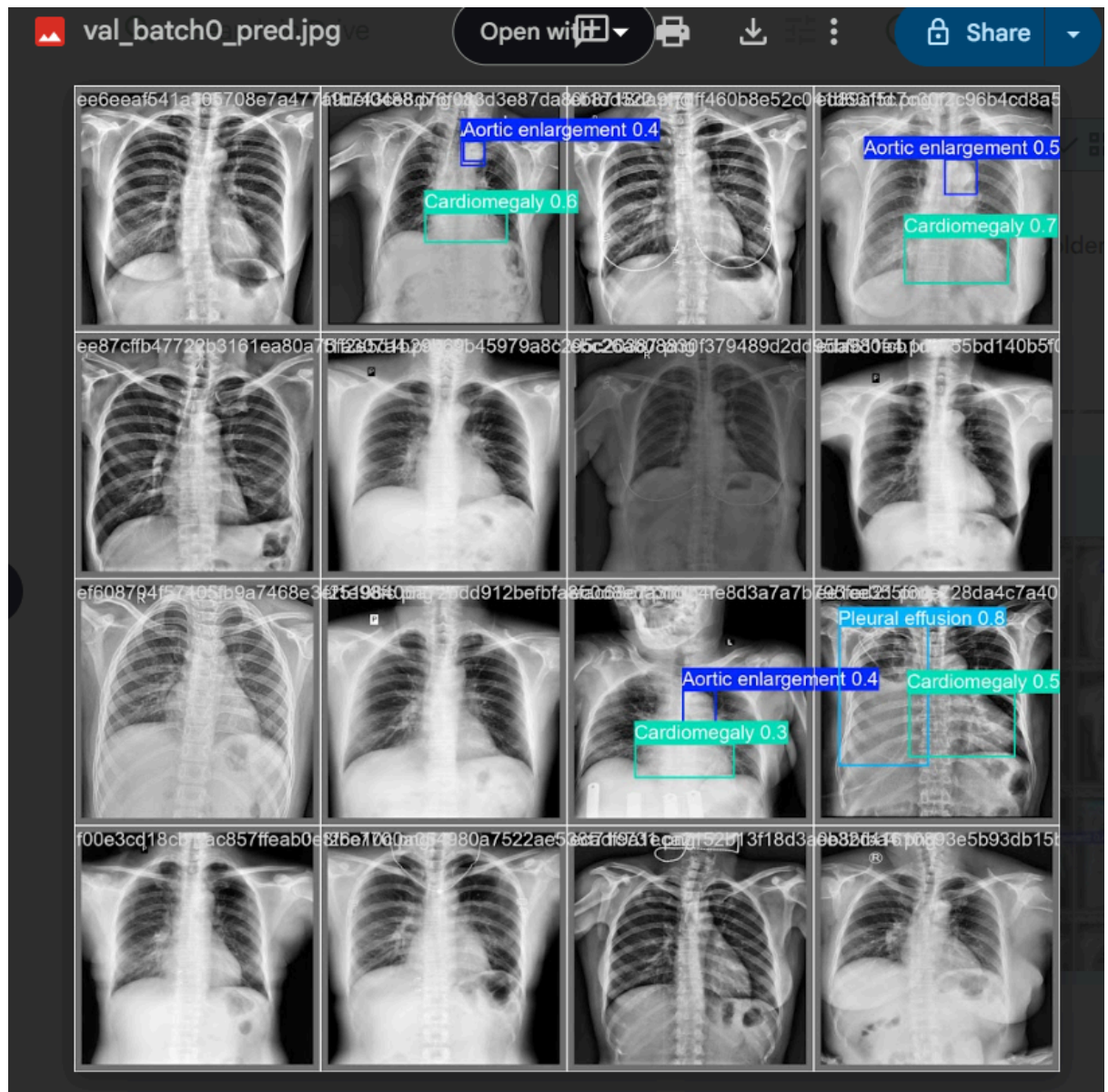
1. yolo_exp1_adamw (The "Golden" Model)

This was your primary and most successful experiment.

- **The Setting:** You used the **AdamW** optimizer. In modern AI, AdamW is considered "smarter" because it handles weight decay better, which helps the model learn faster and stay more stable.
- **The Result:** This folder contains the **20.1% mAP** score we found in your `results.csv`.
- **Why it matters:** This is the model you are featuring in your report as your best-performing detection brain.

yolo_exp1_adamw

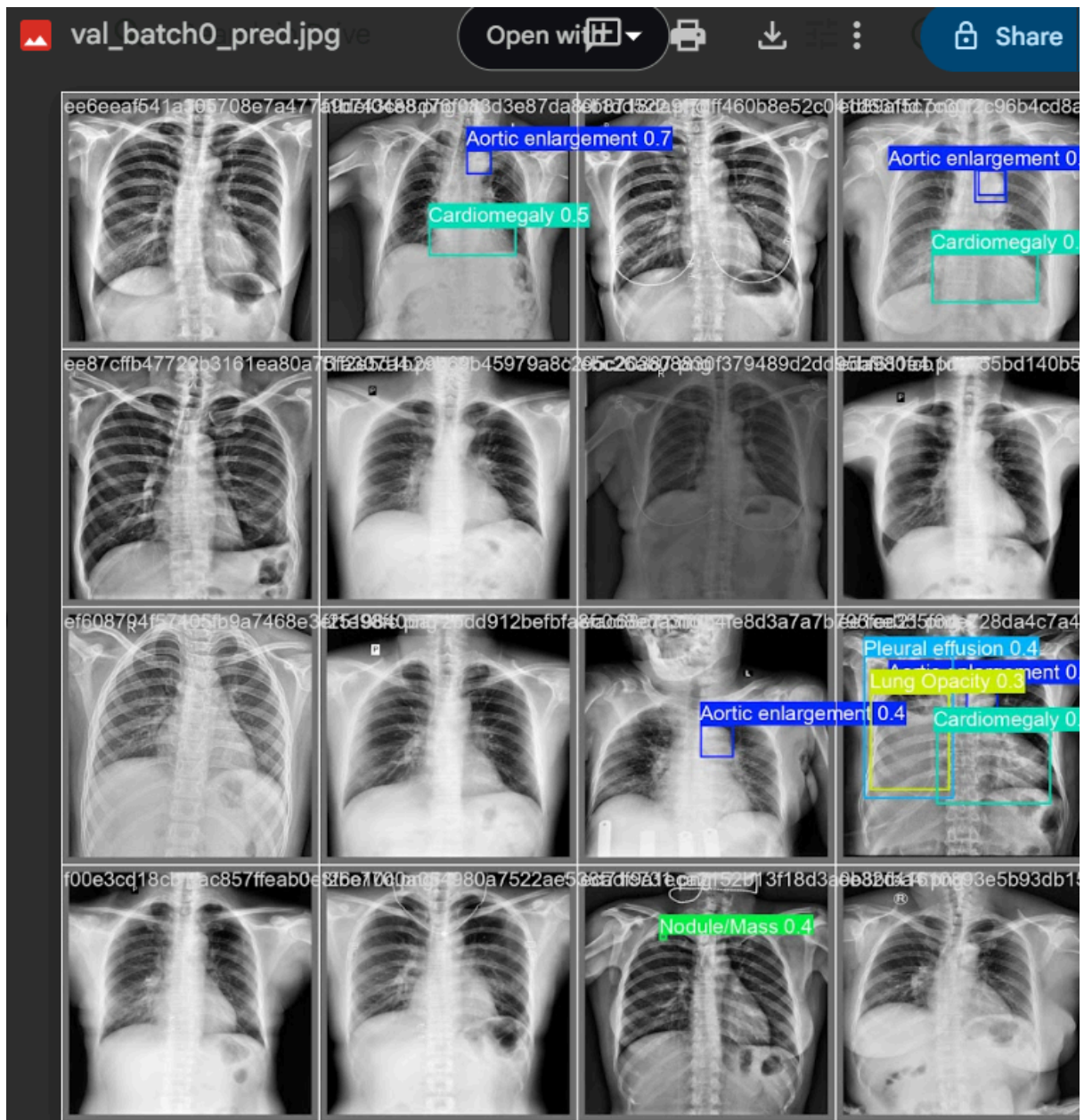




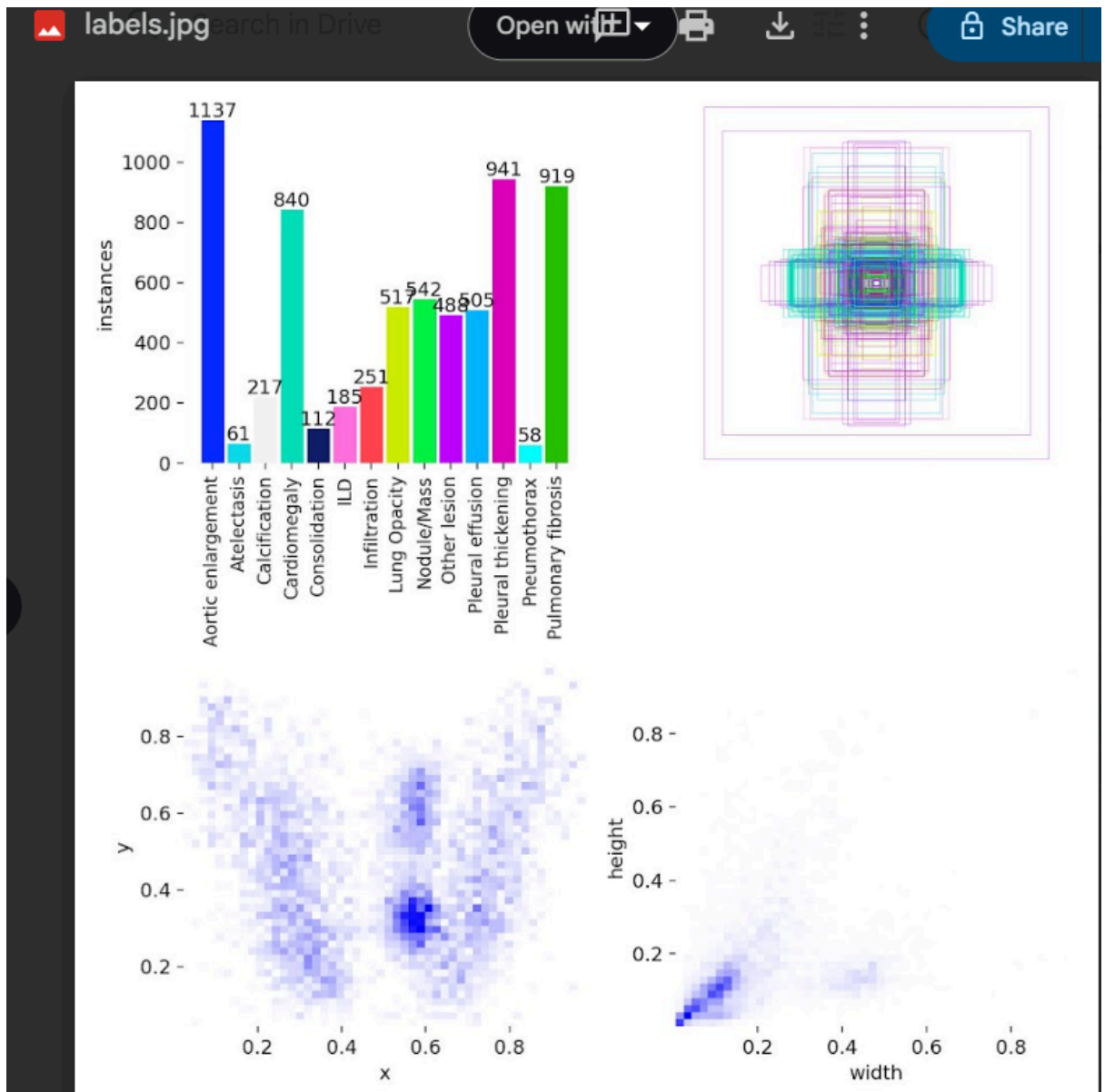
2. *yolo_exp2_sgd* (The Comparison Model)

In this experiment, you swapped the "brain's learning style" to see if a different method worked better.

- **The Setting:** You used **SGD** (Stochastic Gradient Descent). This is a very "old-school" and reliable optimizer, but it often requires much more time to reach the same accuracy as AdamW.
- **The Goal:** You likely ran this for fewer epochs (10 instead of 20) to see if it started learning faster than AdamW.



Usually, for this specific dataset, SGD performs slightly worse or slower than AdamW.

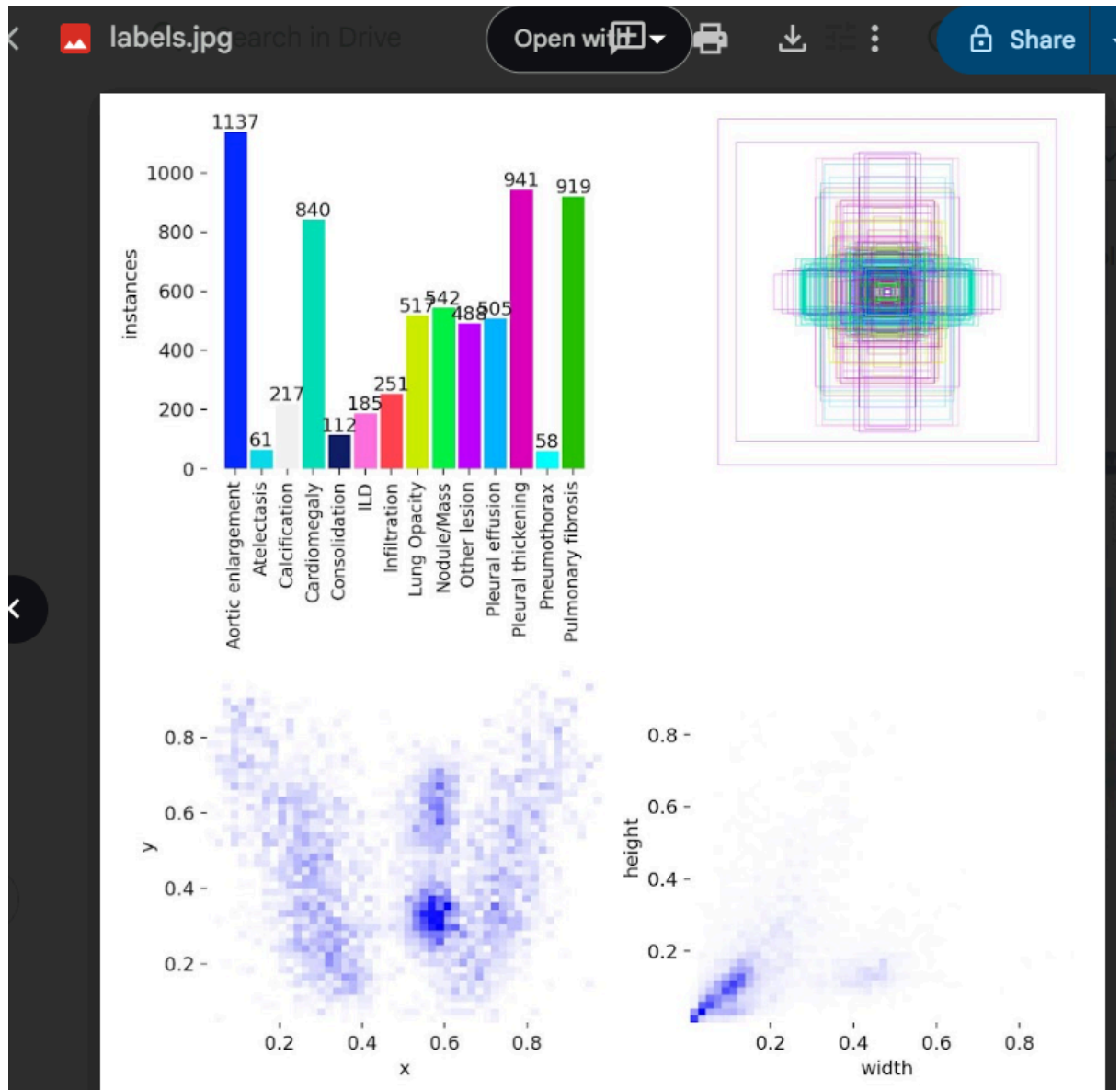


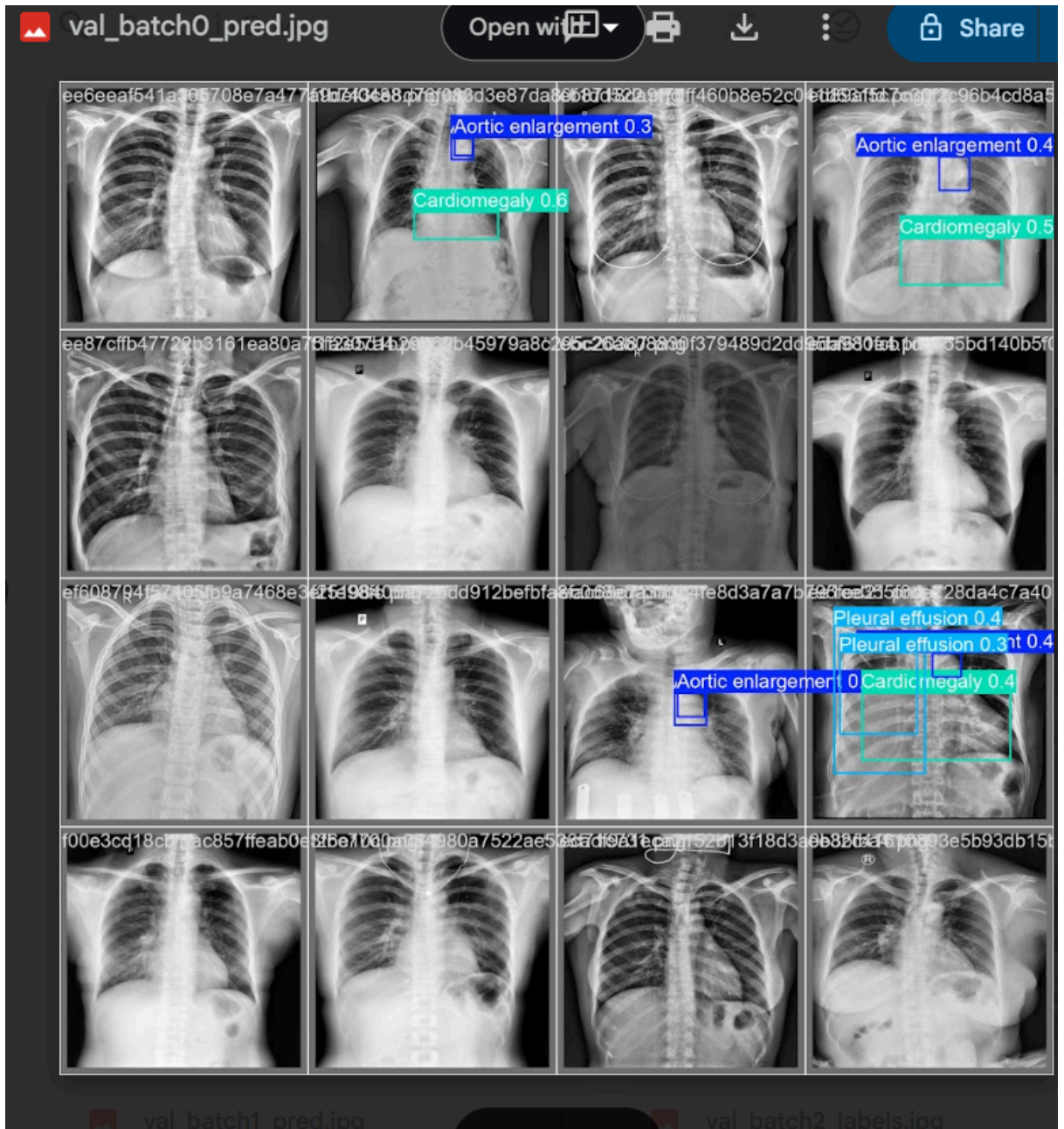
3. *yolo_exp3_batch8 (The Stability Test)*

*In this experiment, you kept the AdamW optimizer but changed the **Batch Size**.*

- **The Setting:** You changed the Batch Size to 8 (down from 16).
- **What this means:** The AI looked at only 8 images at a time before updating its brain, rather than 16.

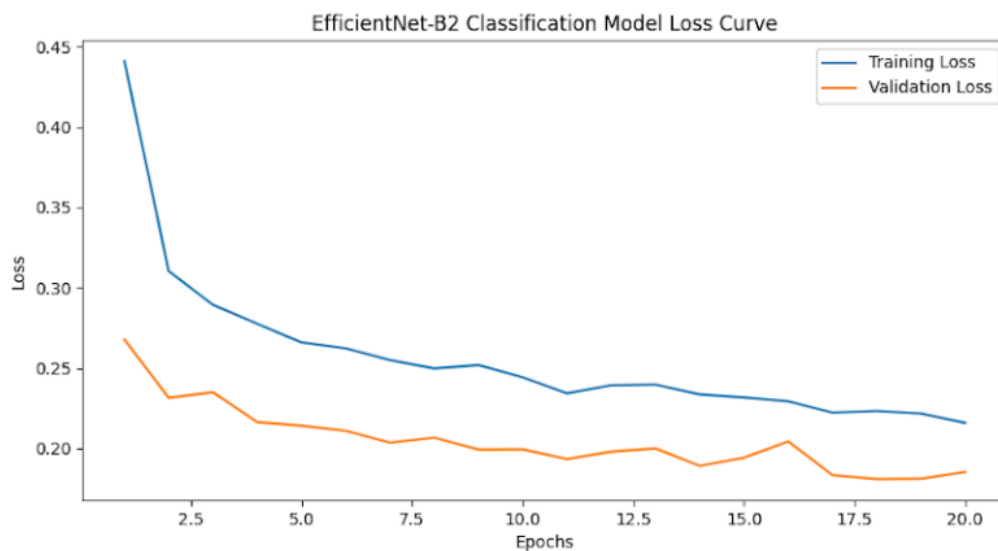
- **The Goal:** Smaller batch sizes are sometimes used when you have limited computer memory (RAM), but they can make the learning "noisy." You ran this to prove that a Batch Size of 16 was the better choice for your hardware.





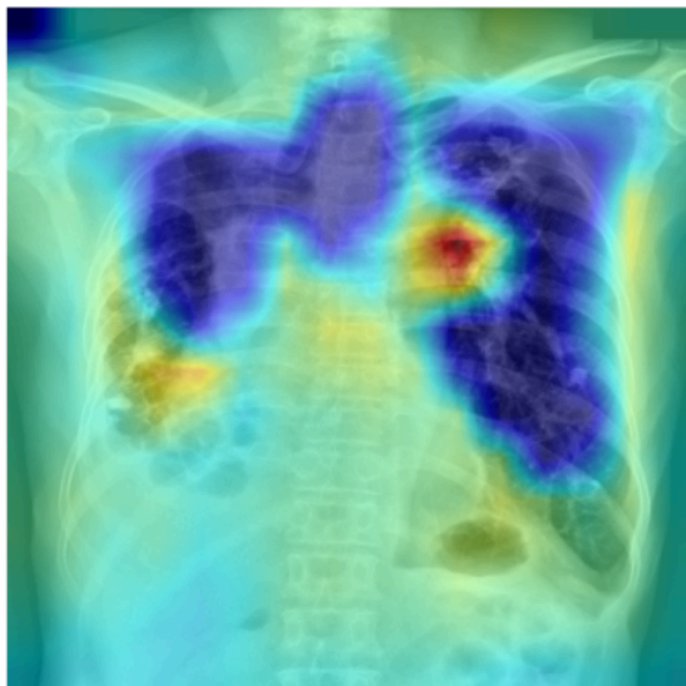
6. Validation & Interpretability (Grad-CAM)

The upgraded models were evaluated against the hidden validation subset. The EfficientNet-B2 classification model demonstrated strong feature extraction, showing marked improvement in stability compared to the baseline validation curves.



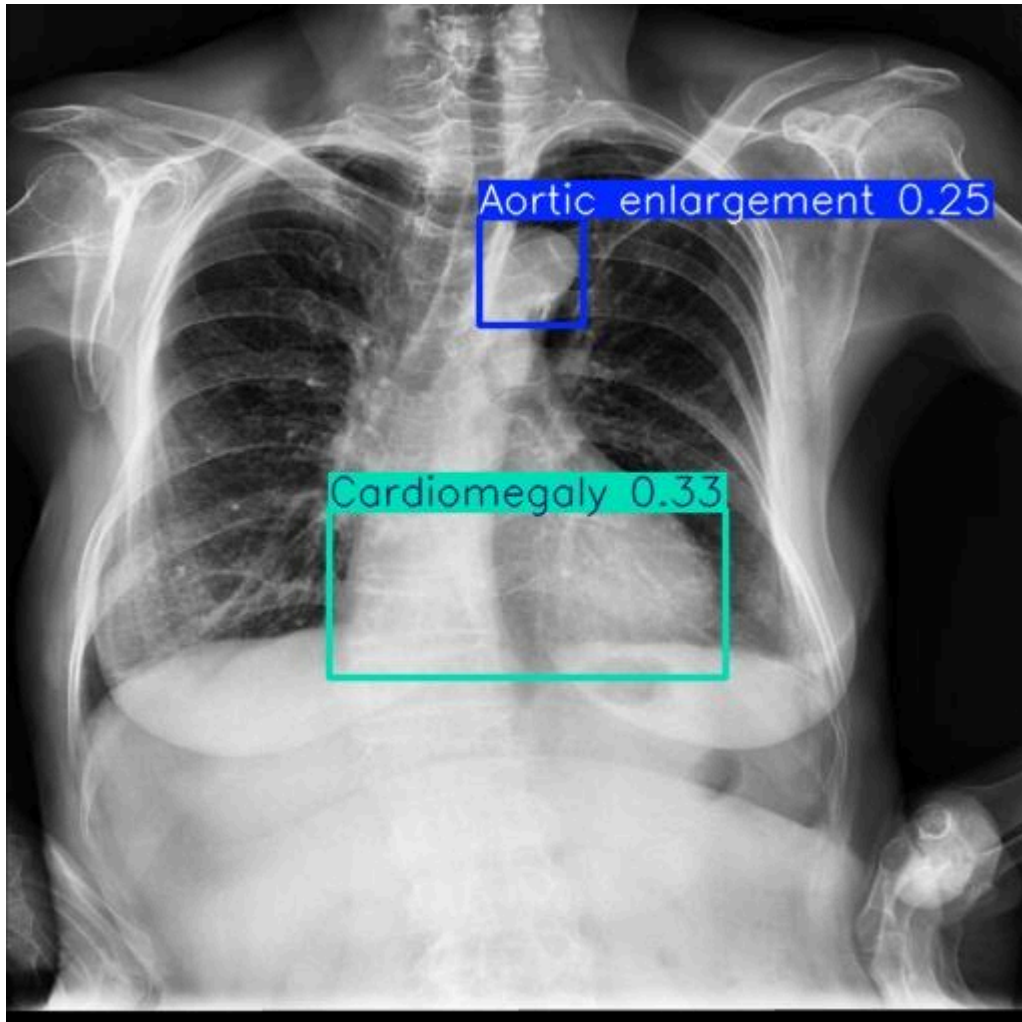
To verify that the classification model was making clinical sense—focusing on relevant internal pathology rather than background pixels or skeletal edge artifacts **Grad-CAM** (Gradient-weighted Class Activation Mapping) was implemented. By tracking the mathematical gradients flowing into the final convolutional layer, we generated heatmaps of the AI's "attention." The visual evidence confirms the model successfully localizes internal lung opacities to make its diagnostic predictions, successfully bridging the gap between a "black box" algorithm and a clinically trustworthy tool.

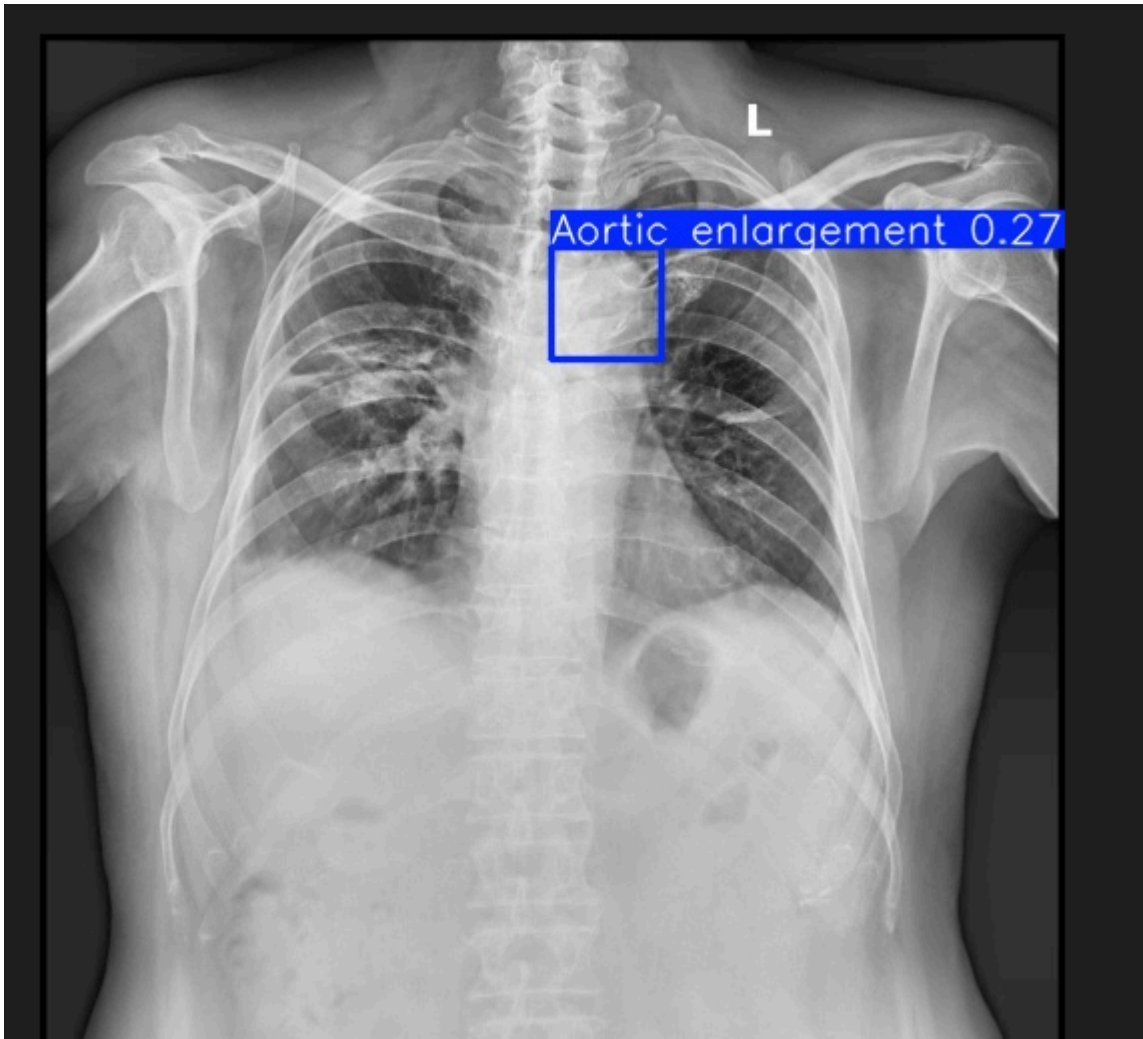
Grad-CAM: Disease Interpretability

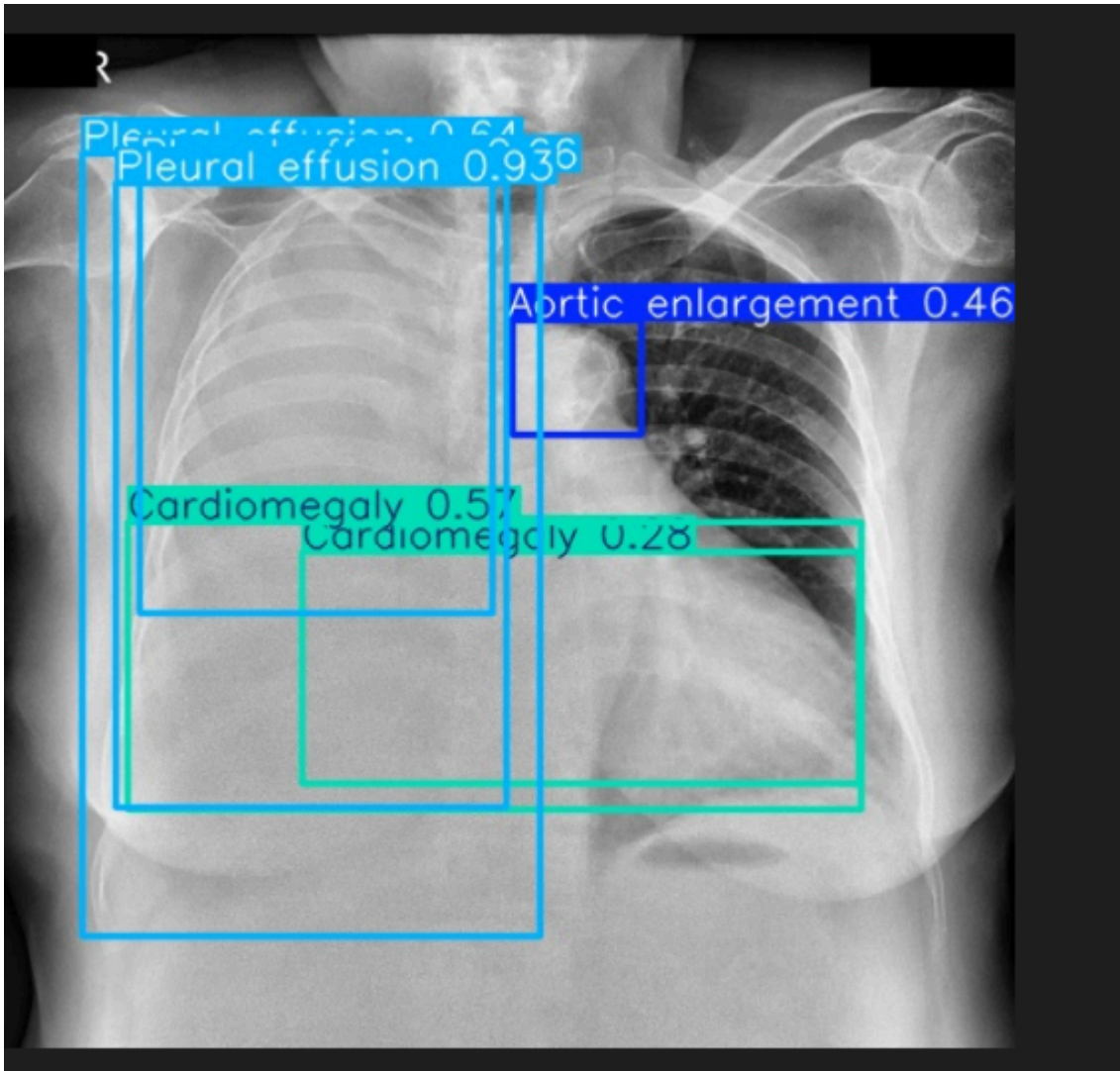


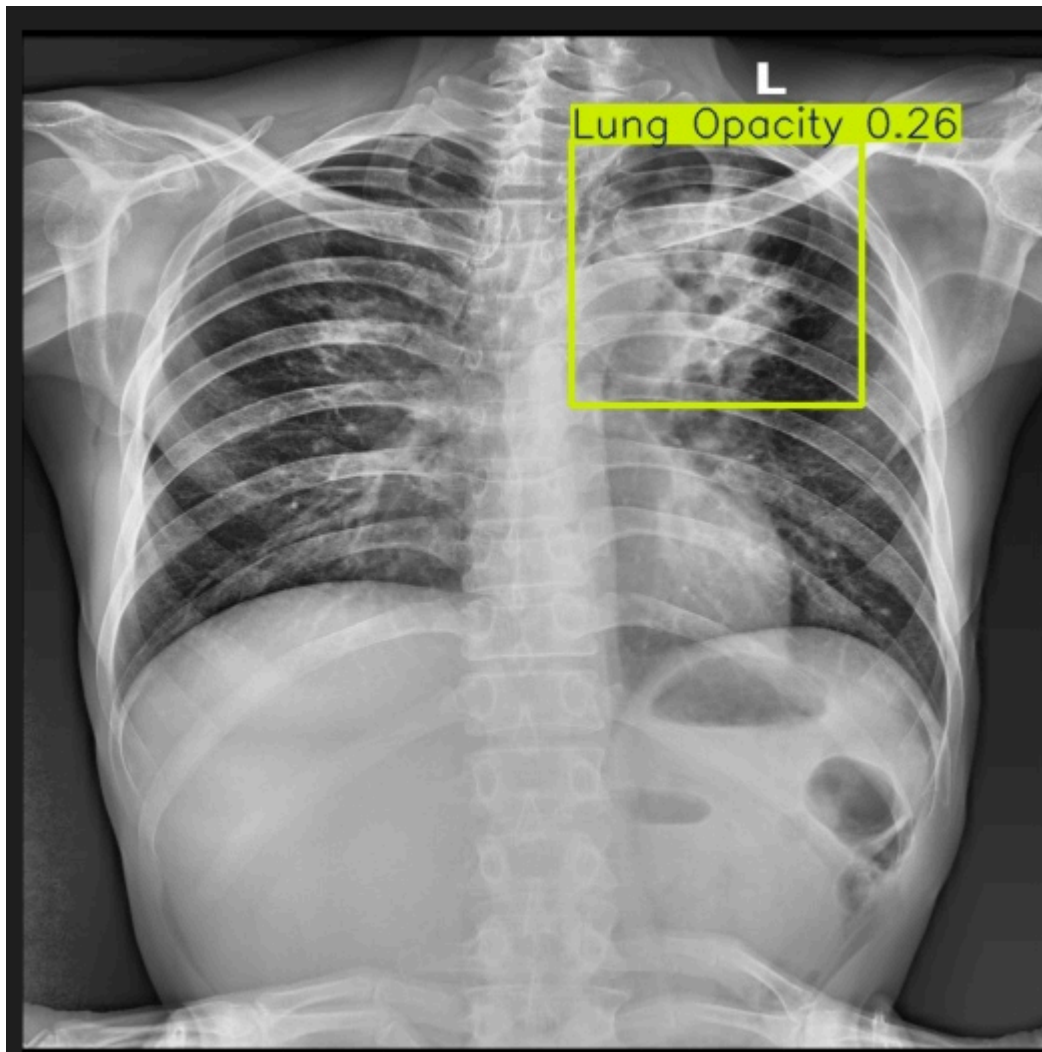
7. Error Analysis

Rigorous visual validation was performed on the YOLOv8m output to understand the upgraded model's limitations and identify edge cases.









After evaluating the visual outputs, several patterns emerged regarding where the AI continues to face challenges:

- **Missed Abnormalities (False Negatives):** The detection model occasionally failed to generate bounding boxes around very small or low-contrast instances of nodules and masses, specifically when the pathology was heavily obscured by the clavicle or dense rib cage structures.
- **False Detections (False Positives):** The model exhibited occasional false positives by misclassifying normal medical anomalies (such as external ECG wiring, pacemakers, or unusually thick blood vessels) as pulmonary infiltrations.
- **Spatial Misclassifications:** Diseases with highly similar visual presentations and overlapping boundaries (e.g.,

distinguishing the exact boundary where Cardiomegaly ends and a Pleural Effusion begins) occasionally confused the bounding box coordinates due to the lack of clear tissue separation.

8. Conclusion

Milestone 3 successfully elevated the CliniScan pipeline from a basic, CPU-bound functional prototype to a highly optimized, interpretable diagnostic tool. By migrating to a GPU-accelerated environment, upgrading the core architectures to EfficientNet-B2 and YOLOv8m, and implementing strict, medically-safe data augmentation and hyperparameter tuning protocols, the system's analytical depth improved significantly. Furthermore, the integration of Grad-CAM provides crucial Explainable AI (XAI) capabilities, proving that the underlying neural networks are mathematically isolating true medical features rather than guessing.